



SHOOLINI UNIVERSITY

Faculty of Engineering & Technology
Department of Computer Science & Engineering

Digital Image Processing

Course Code: CSU2543

Practical 4: Gamma Correction & Power Law Transformations

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Programme: B.Tech CSE (Cybersecurity)
Semester: 8th
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Reference Text: Gonzalez & Woods, *Digital Image Processing*, 3rd Ed.

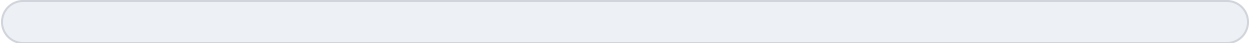
Academic Session 2025–2026

Practical 4: Gamma Correction & Power Law Transformations

Objective

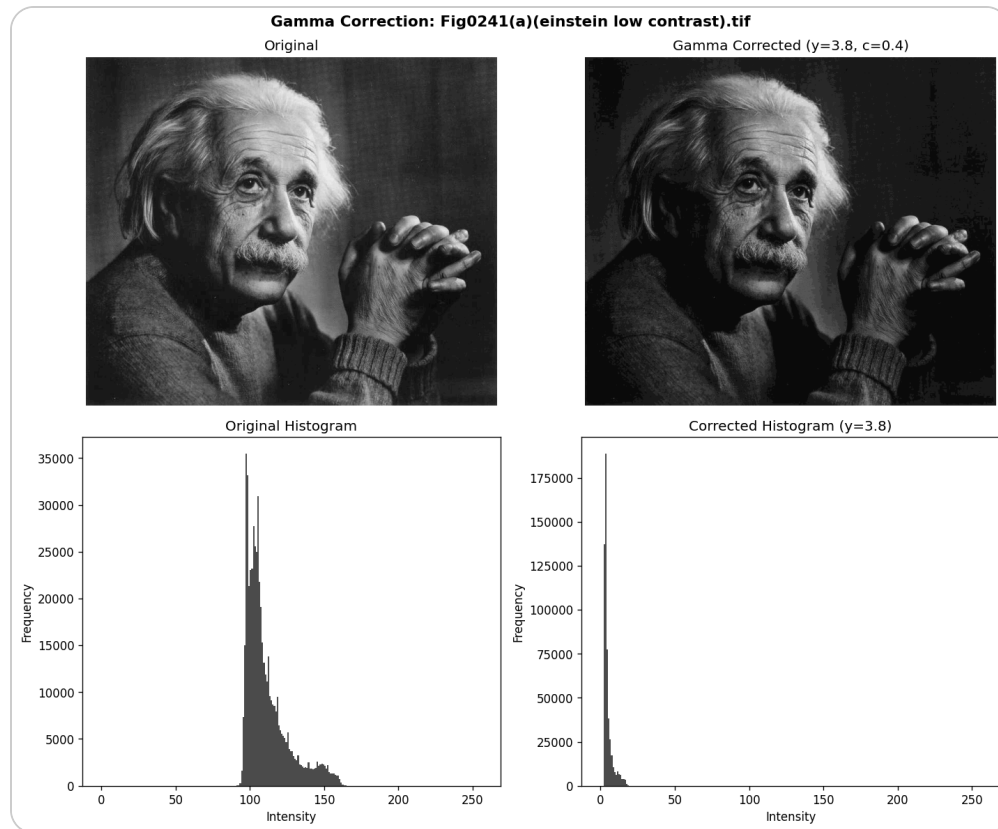
To understand and implement power law (gamma) transformations for image contrast enhancement.

Theory: The power law transformation $s = c \cdot r^\gamma$ is one of the most important intensity transformations.

- $\gamma < 1$: Enhances dark regions (brightening)
 - $\gamma > 1$: Compresses bright regions (darkening)
 - $\gamma = 1, c = 1$: Identity (no change)
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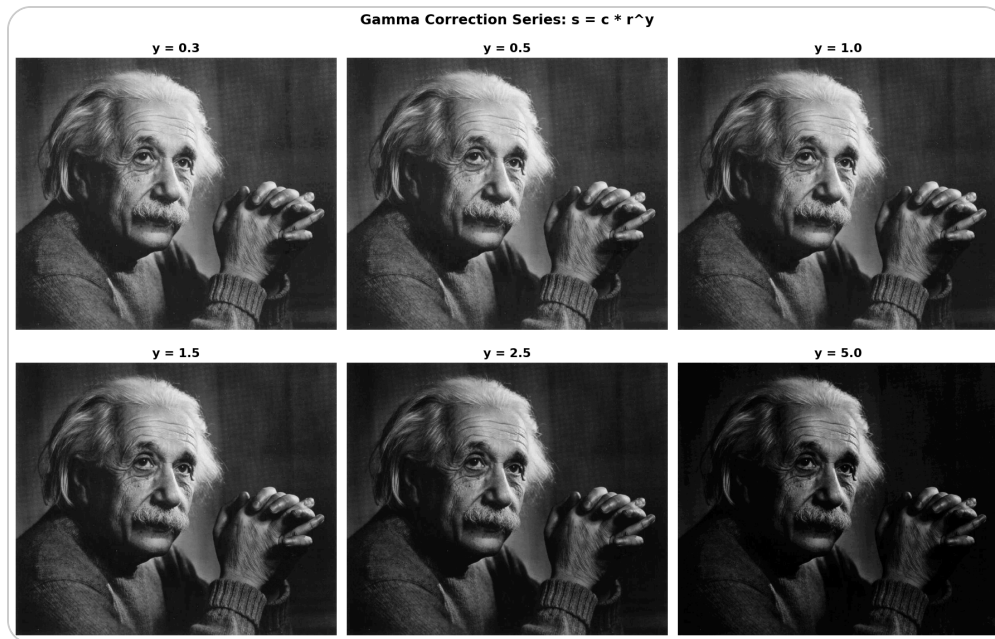
Part 1: Apply Gamma Correction

Adjust the gamma slider and see the transformation result with histogram comparison.



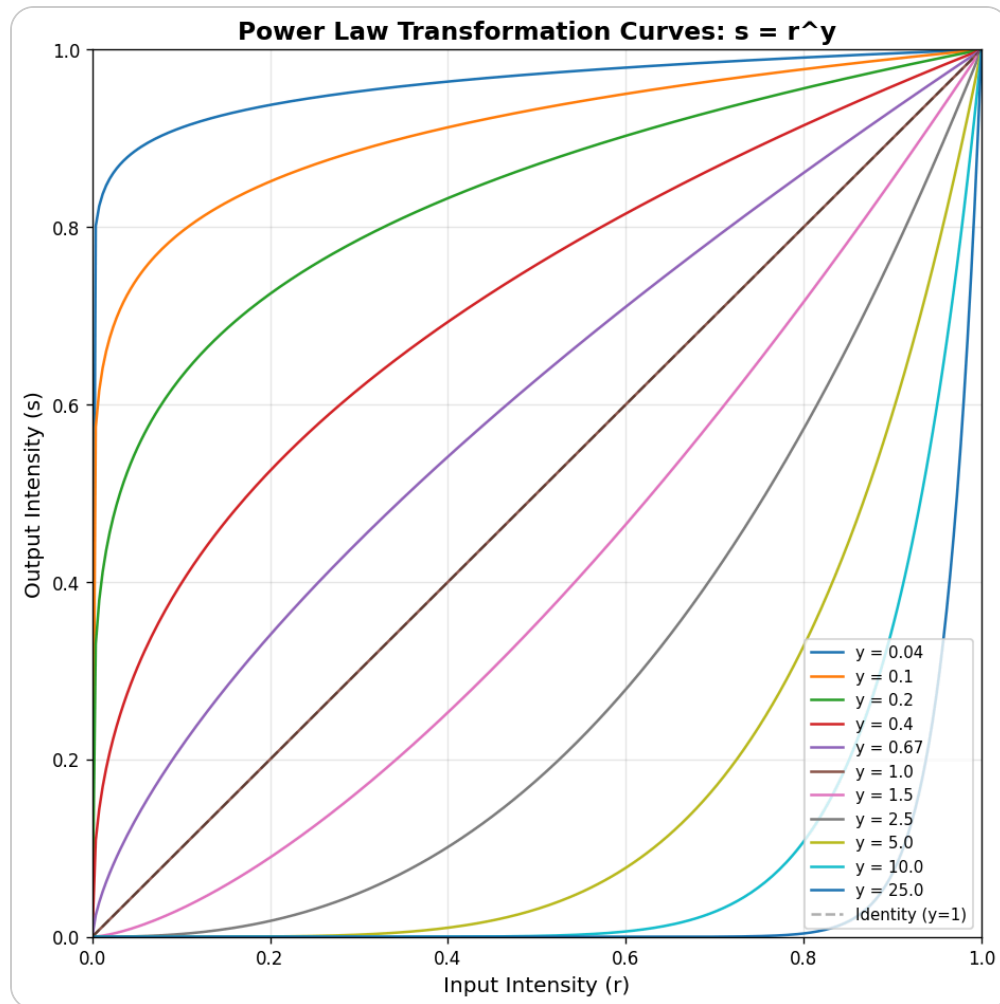
Part 2: Gamma Series Comparison

Compare the effect of multiple gamma values (0.3, 0.5, 1.0, 1.5, 2.5, 5.0) on the selected image.



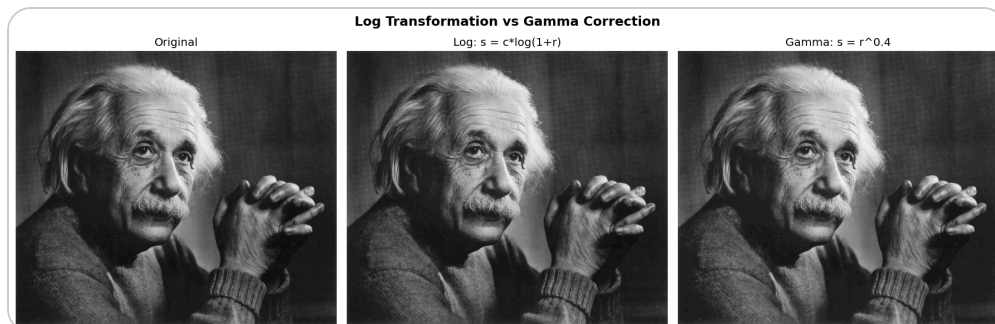
Transformation Curves

Visualize how different γ values map input intensities to output intensities.



Part 3: Log Transformation

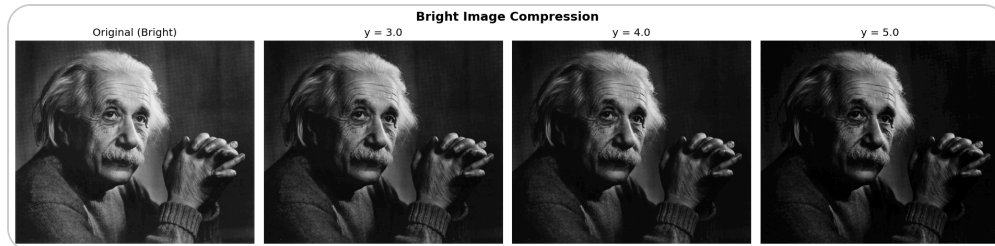
Apply $s = c \cdot \log(1 + r)$ and compare with gamma correction ($\gamma = 0.4$).



Part 4: Contrast Enhancement

Enhance dark images with $\gamma < 1$ or compress bright images with $\gamma > 1$.

☐ Dark Image ($\gamma = 0.3, 0.4, 0.6$) ☒ Bright Image ($\gamma = 3.0, 4.0, 5.0$)



Analysis Questions

1. For a dark image, which gamma value produces the best visual enhancement? Why does $\gamma < 1$ brighten dark regions?
2. Compare log transformation with $\gamma = 0.4$ on the same image. Which produces better results and why?
3. How does gamma correction relate to display devices? If a monitor has $\gamma = 2.5$, what pre-correction gamma should be applied?